

$$F(x, y) = x + y$$

$$x, y \in \mathbb{N}$$

$$D_f \in \mathbb{N} \times \mathbb{N}$$

$$(R_f \in \mathbb{N})$$

$$A = \{1, 2, 3\}$$

$$B = \{2, 4, 6\}$$

$$A \times B = \{(1, 2), (1, 4), (1, 6), (2, 2), (2, 4), (2, 6), (3, 2), (3, 4), (3, 6)\}$$

$$F(g(x))$$

$$F \circ g$$

$$\begin{array}{ccc} x \in \mathbb{N} & g(x) \in \mathbb{N} & \\ D_g & \searrow & R_g \\ D_f & \swarrow & R_f \in \mathbb{N} \end{array}$$

$$F(x)(y)$$

$$\begin{array}{l} F((x: \text{Int}) (\text{Int} \Rightarrow \text{Int}) \{ \\ \quad (y: \text{Int}) \Rightarrow x + y \\ \}) \end{array}$$

$$\begin{array}{ccc} & x & \\ F(x) & \boxed{y} & D_f \in \mathbb{N} \\ & & R_f \in \mathbb{N} \times \mathbb{N} \\ & & \begin{array}{cc} y & x + y \end{array} \end{array}$$

